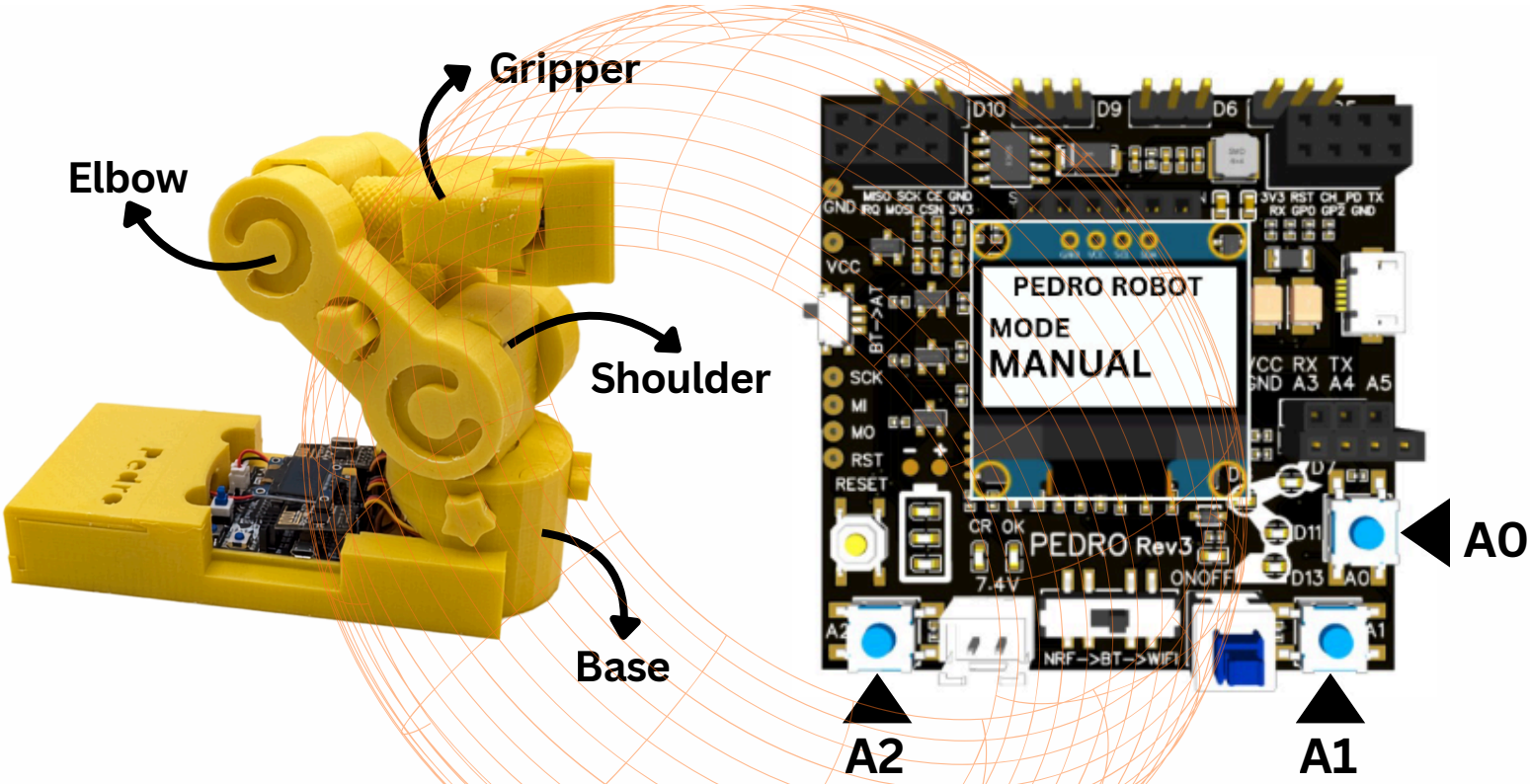
BEGINNER

pedro

90 min

STEM Program

Lesson #4: Manual Mode



Complete this lesson to earn your Pedro Badge



Add it to your Pedro Passport to track your STEM journey

Your mission: Learn how to control Pedro robot by using the manual mode.

1. Learning Goals

In the previous lesson (Lesson #3: Assembly), you learned how to assemble the Pedro robot.

In this new lesson, you will discover how one of Pedro robot's seven modes works: Manual Mode.

At the end of this lesson, students will be able to unlock their new badge and complete their learning passport.



2. Teacher Notes

- No robotics experience is required.
- Duration: ~90 minutes
- Required matériels:
  - **1 assembled Pedro robot** per student or per group (recommended: groups of 2 students).

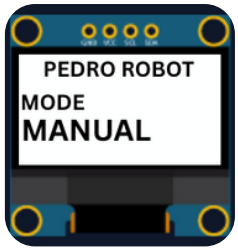
3. Programming the Pedro board

Skip this chapter if your Pedro robot is already programmed; otherwise, review Chapter 4 of Lesson #3.

4. Manual Mode

How does it work

After programming the Pedro board with the **PedroFirmware** program, the board automatically restarts in Manual Mode. The Pedro robot's screen displays **mode MANUAL** by default.

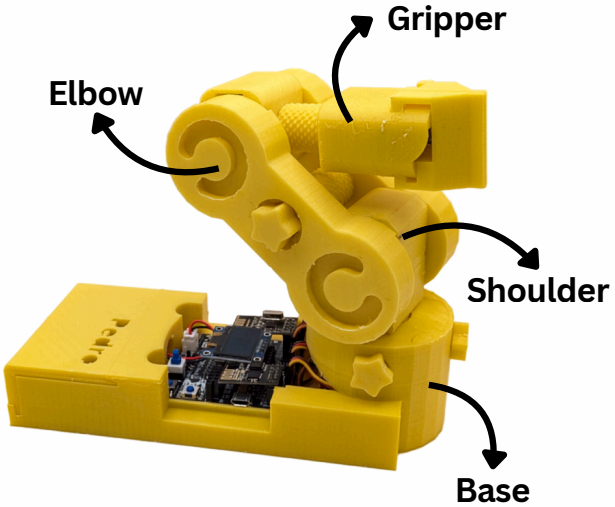


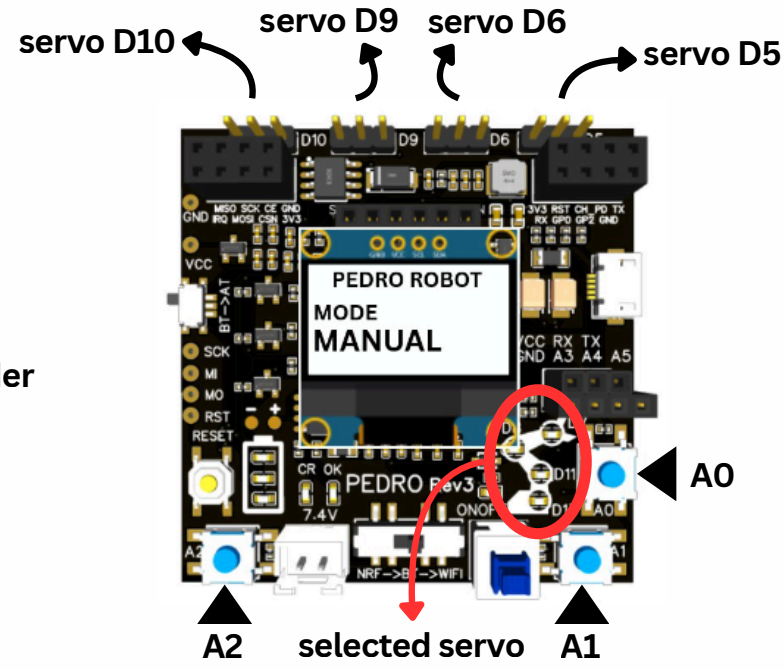
This mode lets you control the robot directly using the buttons on the board:

- **A0** : selects the robot section (**Base**, **Shoulder**, **Elbow**, **Gripper**).
- **A1** and **A2** : rotate the selected section.

Each selected Pedro section is indicated by LEDs:

- LED **D13** → **Base** selected (servo **D5**)
- LED **D11** → **Shoulder** selected (servo **D6**)
- LED **D8** → **Elbow** selected (servo **D9**)
- LED **D7** → **Gripper** selected (servo **D10**)





5. Manual Mode: Test

Make sure you complete each of the following steps:

Base test:

- ☐ Turn on the Pedro robot.
- ☐ Check that **LED D13** is **ON**.
- ☐ Press **A1** and check that the **base** rotates to its limit.
- ☐ Press **A2** and check that the **base** rotates in the opposite direction.

Shoulder test:

- ☐ Select the **shoulder** by pressing **A0**.
- ☐ Check that **LED D11** is **ON**.
- ☐ Press **A1** and check that the **shoulder** rotates to its limit.
- ☐ Press **A2** and check that the **shoulder** rotates in the opposite direction.

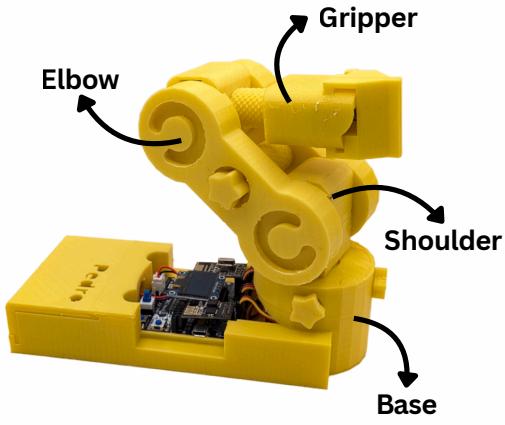
Elbow test:

- ☐ Select the **elbow** by pressing **A0**.
- ☐ Check that **LED D8** is **ON**.
- ☐ Press **A1** and check that the **elbow** rotates to its limit.
- ☐ Press **A2** and check that the **elbow** rotates in the opposite direction.

Gripper test:

- ☐ Select the **gripper** by pressing **A0**.
- ☐ Check that **LED D7** is **ON**.
- ☐ Test opening and closing the **gripper** using buttons **A1** and **A2**.

If a section remains stuck while rotating, check the servo wiring and, if necessary, the gear position.



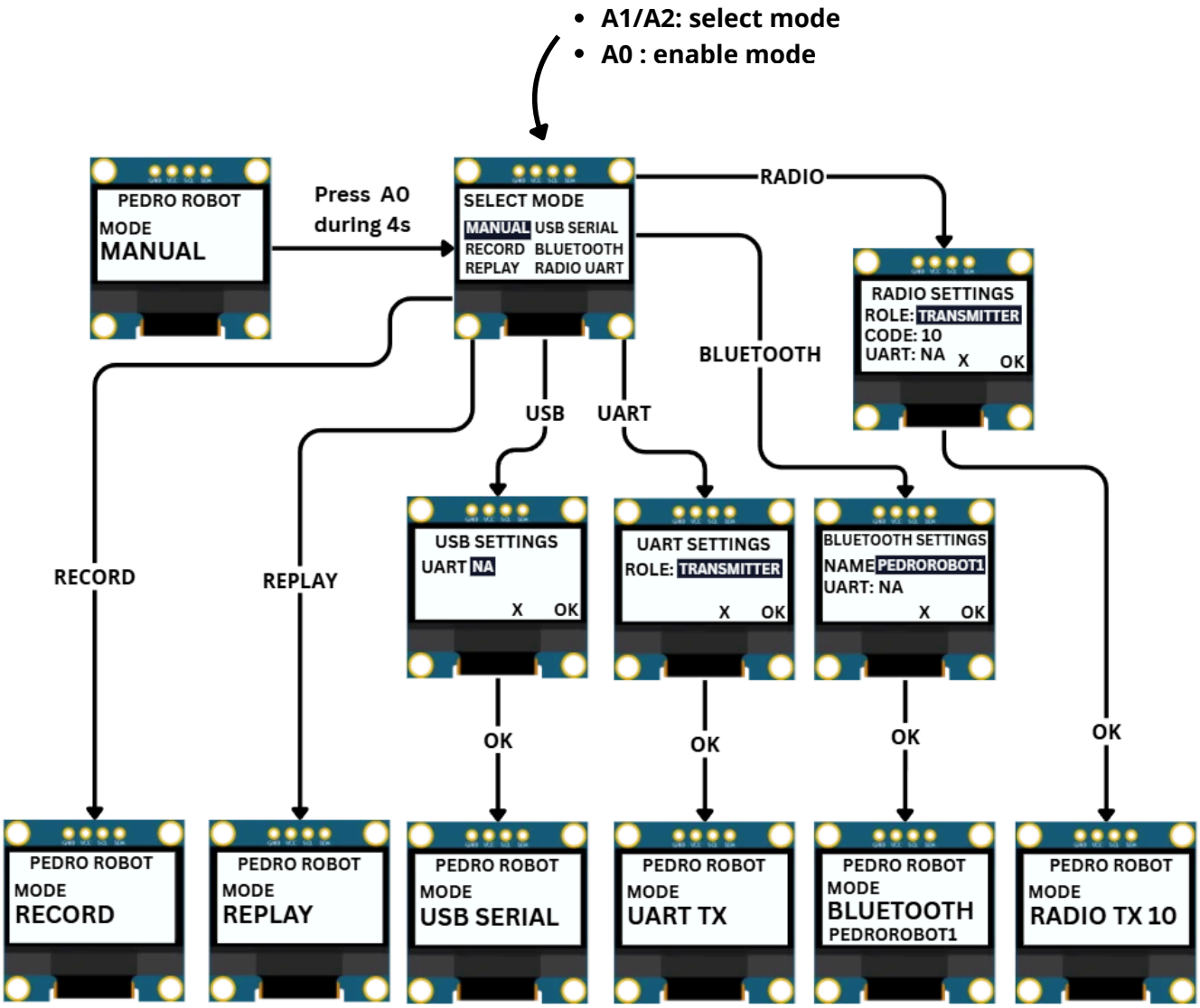
6. Control Modes

As seen in previous lessons, the Pedro robot has seven control modes.

Each mode allows you to control Pedro in a different way:

1. **Manual Mode:** direct control using the buttons.
2. **Record Mode:** records movements.
3. **Replay Mode:** plays back recorded movements.
4. **USB Mode:** computer control via USB cable.
5. **Bluetooth Mode:** control via smartphone.
6. **Radio Mode:** remote control via radio link.
7. **UART Mode:** wired control via RX/TX connection.

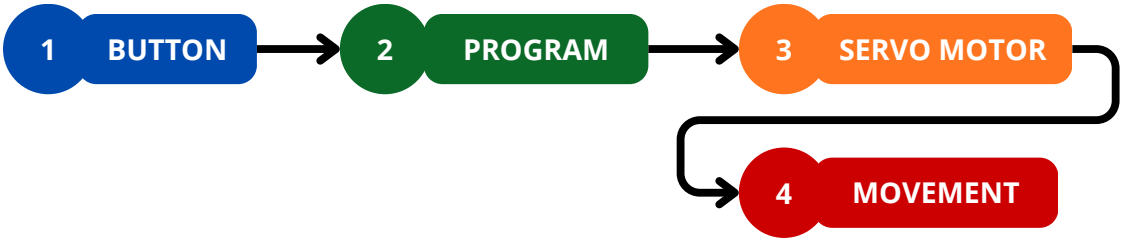
You can navigate through Pedro's different modes using the diagram below:



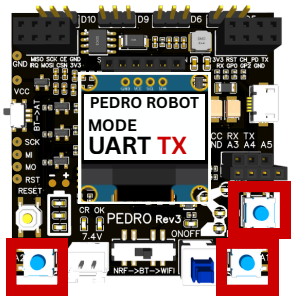
7. Manual Mode: Processes

In robotics, a process is a defined step in how a robot operates. Each process has a specific function.

The processes describing Pedro robot's Manual Mode can be represented as follows:

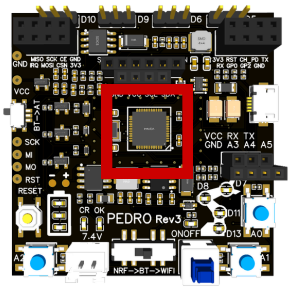


1 BUTTON



Select the servo motor using button **A0** and control it using buttons **A1** and **A2**.

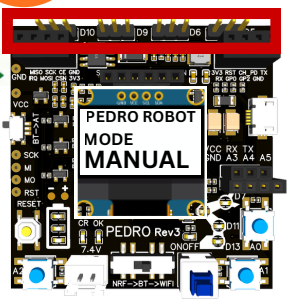
2 PROGRAM



The **microcontroller** receives the button commands and sends a **PWM signal** to the selected servo motor.

PWM Signal

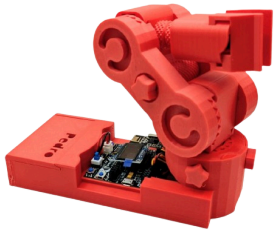
3 SERVO MOTOR



The selected servo motor receives the **PWM signal** and rotates in one direction according to button **A1** or **A2**.

4 MOVEMENT

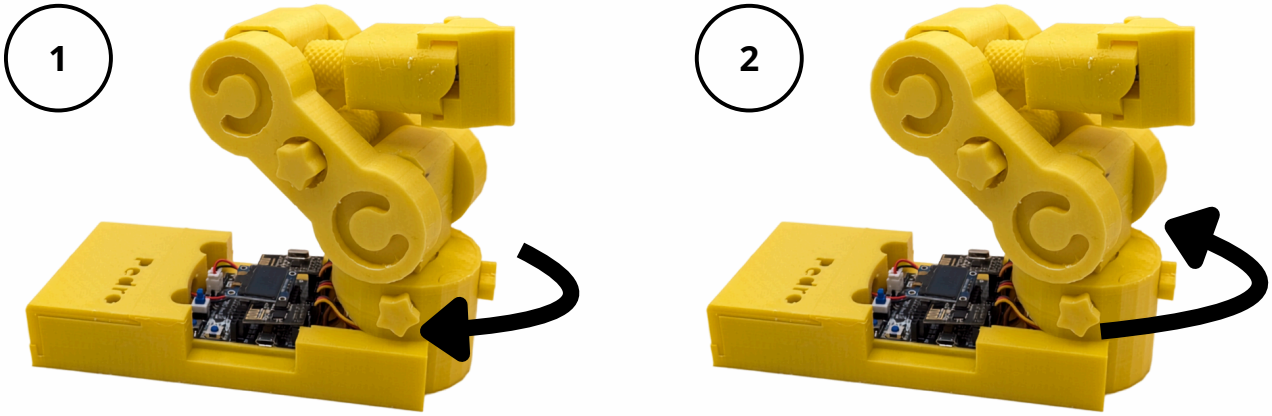
The rotation of the selected servo motor causes the robot section to move (base, elbow, shoulder, or gripper)



8. Exercises

Base:

1. Select the base by using the button **A0**. Which LED is ON? _____
2. By using **A1** and **A2** rotate the base as shown in the picture.



3. Which button rotates the base in direction 1? _____
4. Which button rotates the base in direction 2? _____
5. Complete the program below using these words:
   HIGH, LOW, RIGHT, LEFT, A1, A2, D5, D6, D7, D8, D9, D10, D11, D13

```
// Check the LED state
IF (digitalRead(D...) == .........) THEN {
 OBSERVATION → LED D... (base) is ON
} SINON SI (digitalRead(D...) == .........) THEN {
 OBSERVATION → LED D... (base) is OFF
}

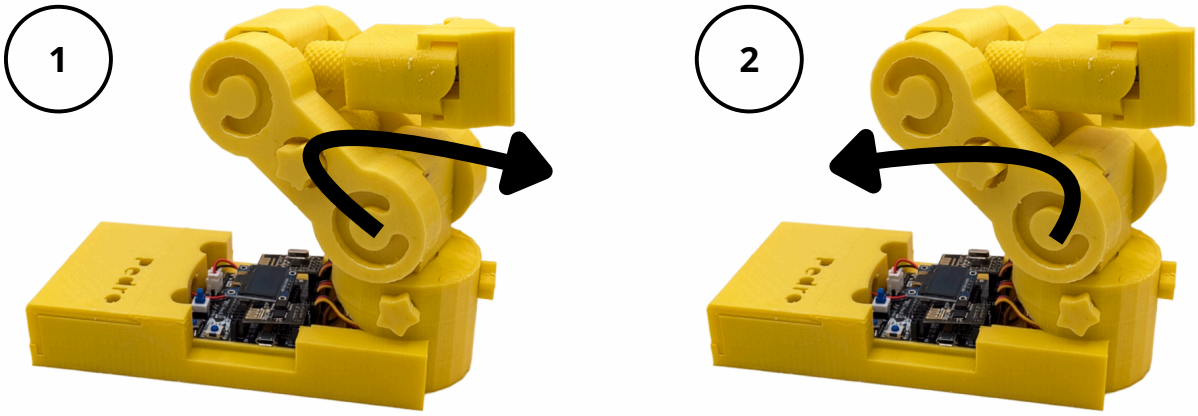
// Check the button state
IF (digitalRead(A...) == .........) THEN {
  ACTION → The servo motor D... rotates to the ..........
} ELSE IF (digitalRead(A...) == .........) THEN {
  ACTION → The servo motor D... rotates to the ..........
} ELSE {
  ACTION → The servo motor stops
}
```

digitalRead is a function that reads the state of a button or LED:

- HIGH : button pressed or LED ON.
- LOW : button released or LED OFF.

Shoulder

1. Select the shoulder by using the button **A0**.
2. Which LED is ON? _____
3. By using **A1** and **A2** rotate the shoulder as shown in the picture.



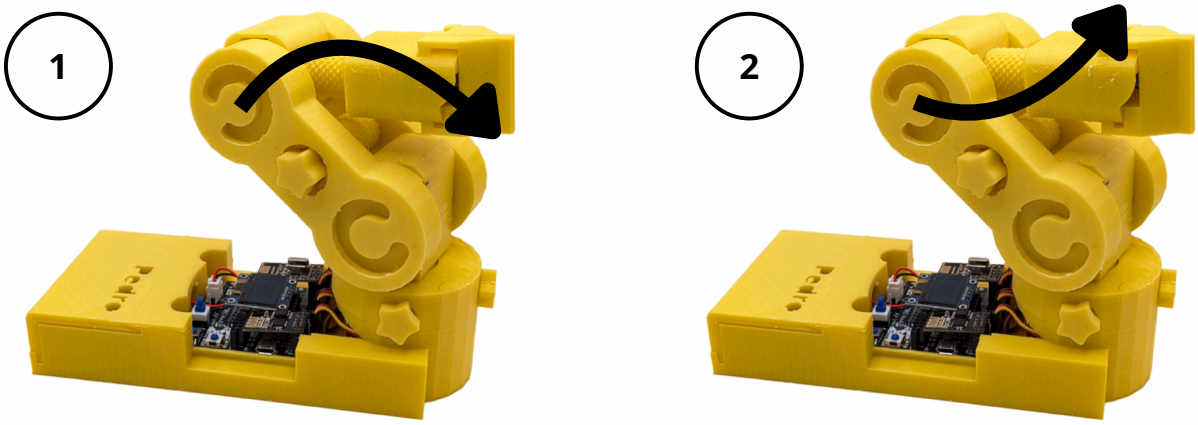
3. Which button rotates the **shoulder** in direction 1? _____
4. Which button rotates the **shoulder** in direction 2 ? _____
5. Complete the program below using these words:
   HIGH, LOW, FORWARD, BACKWARD, A1, A2, D5, D6, D7, D8, D9, D10, D11, D13

```
// Check the LED state
IF (digitalRead(....) == .........) THEN {
 OBSERVATION → LED .... (shoulder) is ON
} SINON SI (digitalRead(D...) == .........) THEN {
 OBSERVATION → LED .... (shoulder) is OFF
}

// Check the button state
IF (digitalRead(....) == .........) THEN {
  ACTION → The servo motor .... rotates ..........
} ELSE IF (digitalRead(A...) == .........) THEN {
  ACTION → The servo motor .... rotates ..........
} ELSE {
  ACTION → The servo motor stops
}
```

Elbow

1. Select the elbow by using the button **A0**.
2. Which LED is ON? _____
3. By using **A1** and **A2** rotate the elbow as shown in the picture.



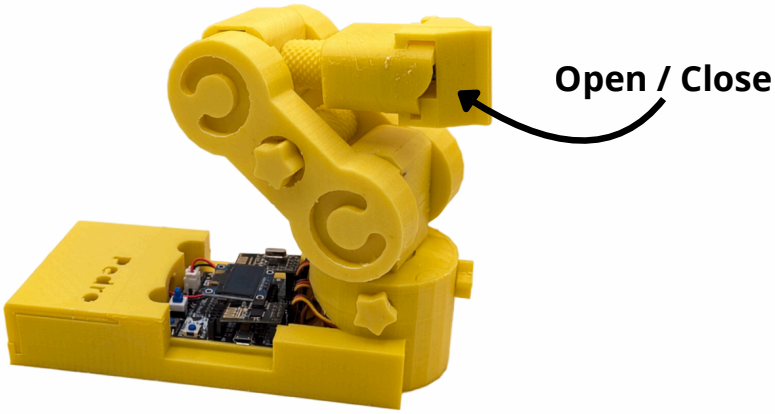
3. Which button rotates the elbow in direction 1? _____
4. Which button rotates the elbow in direction 2? _____
5. Complete the program below using these words:
   HIGH, LOW, UP, DOWN, A1, A2, D5, D6, D7, D8, D9, D10, D11, D13

```
// Check the LED state
IF (digitalRead(....) == .........) THEN {
 OBSERVATION → LED .... (elbow) is ON
} SINON SI (digitalRead(D...) == .........) THEN {
 OBSERVATION → LED .... (elbow) is OFF
}

// Check the button state
IF (digitalRead(....) == .........) THEN {
  ACTION → The servo motor .... rotates ..........
} ELSE IF (digitalRead(A...) == .........) THEN {
  ACTION → The servo motor .... rotates ..........
} ELSE {
  ACTION → The servo motor stops
}
```

Gripper

1. Select the gripper by using the button **A0**.
2. Which LED is ON? _____
3. By using **A1** and **A2** open and close the gripper as shown in the picture.



3. Which button opens the gripper? _____
4. Which button closes the gripper? _____
5. Complete the program below using these words:
   HIGH, LOW, OPEN, CLOSE, A1, A2, D5, D6, D7, D8, D9, D10, D11, D13

```
// Check the LED state
IF (digitalRead(....) == .........) THEN {
 OBSERVATION → LED .... (gripper) is ON
} SINON SI (digitalRead(D...) == .........) THEN {
 OBSERVATION → LED .... (gripper) is OFF
}

// Check the button state
IF (digitalRead(....) == .........) THEN {
  ACTION → The servo motor .......... the gripper
} ELSE IF (digitalRead(A...) == .........) THEN {
  ACTION → The servo motor .......... the gripper
} ELSE {
  ACTION → The servo motor stops
}
```

Quiz

1. What is the default operating mode when Pedro starts?

- [ ] Bluetooth
- [ ] Radio
- [ ] Manual
- [ ] USB Serial

2. Which button selects the active servo?

- [ ] A0
- [ ] A1
- [ ] A2
- [ ] Reset

3. Which buttons move the selected servo?

- [ ] A0
- [ ] A1
- [ ] None
- [ ] A2

4. Which LED is ON when the elbow is selected?

- [ ] D7
- [ ] D13
- [ ] D8
- [ ] D11

5. What component acts as Pedro's brain?

- [ ] Servo motor
- [ ] OLED screen
- [ ] Microcontroller
- [ ] Battery

6. Which signal is sent to the servo motor by the microcontroller?

- [ ] Bluetooth
- [ ] PWM
- [ ] Wi-Fi
- [ ] Radio

7. Which process is triggered immediately after pressing buttons A1 or A2?

- [ ] The servo rotates.
- [ ] The microcontroller reads the button state.
- [ ] The OLED turns off.
- [ ] The battery charges.

8. Which processes describe how Manual Mode works?

- [ ] Movement → Servo motor → Button → Program
- [ ] Button → Program → Servo motor → Movement
- [ ] Battery → Servo motor → Movement → Button
- [ ] Screen → Program → Battery → Movement

9. What is the microcontroller's main role?

- [ ] Prints the robot.
- [ ] Reads inputs and executes the program.
- [ ] Recharge the battery.
- [ ] Prevent the gears from rotating.

10. What is the purpose of the digitalRead function?

- [ ] Send a PWM signal.
- [ ] Turn on the LED.
- [ ] Read the state of the LED or button.
- [ ] Turn on the robot.

🎯 Congratulations Mission Completed!

You have just unlocked the Pedro **Manual Mode** badge.
Add it to your passport and see you in the next lesson!

